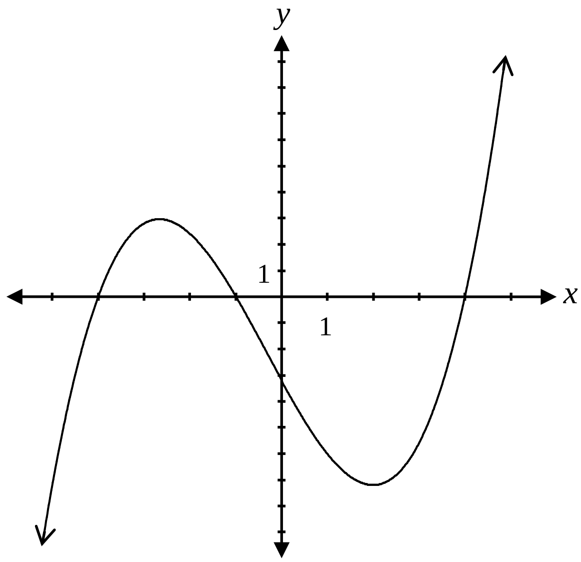
Question 20 — 1 mark

Given the graph of the function of $f(x)$ below, what is the range of $y = |f(x)|$?



a) $y \in$ ¡ b) $y \geq -7$ c) $y \geq 0$ d) $-4 \leq y \leq -1$ or $y \geq 4$

Question 21 — 1 mark

Simplify the following expression:

$$\frac{1}{2}\log_a 36 - \log_a 2$$

a) $\log_a 3$ b) $\log_a 4$ c) $\log_a 9$ d) $\log_a 12$

Question 22 — 1 mark

Given $f(x) = x^2 - x + 2,$ an equation that represents the graph of $f(x)$ shifted 3 units to the right is:

a) $y = (x+3)^2 - (x+3) - 3$

b) $y = (x-3)^2 - (x-3) + 2$

c) $y = (x-3)^2 - x - 2$

d) $y = x^2 - x + 2 - 3$

Question 23 — 1 mark

What is the domain of the function $y = \sqrt{-4x}$?

a) $\{x \in \mathbb{R} \mid x \geq 2\}$

b) $\{x \in \mathbb{R} \mid x \leq 2\}$

c) $\{x \in \mathbb{R} \mid x \geq 0\}$

d) $\{x \in \mathbb{R} \mid x \leq 0\}$

Question 24 — 1 mark

Which of the following is true about the two functions below?

$$f(x) = \frac{(x+2)(x-2)}{x-2} \qquad g(x) = \frac{(x-2)(x+1)}{(x+2)(x-2)}$$

a) Both have a point of discontinuity (hole) when $x = 2$.

b) Both have the same vertical asymptote.

c) Both have the same horizontal asymptote.

d) Both have the same y-intercept.

Question 25 **1 mark**

The general solution to the equation $\cos\theta = -\frac{1}{2}$ is:

a) $\left.\begin{array}{l} \theta = \frac{\pi}{3} + 2\pi k \\ \theta = \frac{5\pi}{3} + 2\pi k \end{array}\right\}$ where $k \in \mathrm{I}$

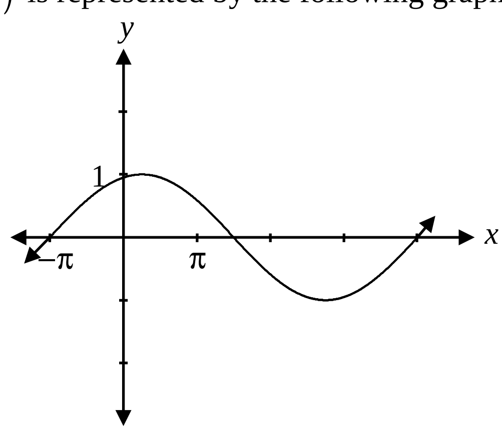
b) $\left.\begin{array}{l} \theta = \frac{\pi}{3} + \pi k \\ \theta = \frac{5\pi}{3} + \pi k \end{array}\right\}$ where $k \in \mathrm{I}$

c) $\left.\begin{array}{l} \theta = \frac{2\pi}{3} + 2\pi k \\ \theta = \frac{4\pi}{3} + 2\pi k \end{array}\right\}$ where $k \in \mathrm{I}$

d) $\left.\begin{array}{l} \theta = \frac{2\pi}{3} + \pi k \\ \theta = \frac{4\pi}{3} + \pi k \end{array}\right\}$ where $k \in \mathrm{I}$

Question 26 **1 mark**

If the equation $y = \sin\left(b\left(x+\pi\right)\right)$ is represented by the following graph, what is the value of b?



a) $\frac{2}{5}$

b) $\frac{5}{2}$

c) $\frac{2\pi}{5}$

d) 5π

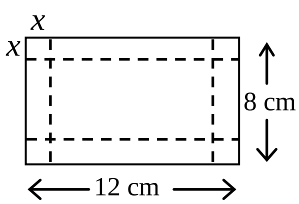
Question 27 — 1 mark

Which of the following is closest to the value of $\log_2 40 + \log_5 125$?

a) 3 b) 8 c) 10 d) 45

Question 28 — 1 mark

A sheet of paper 12 cm long and 8 cm wide is used to make a box with no lid. Equal squares of side length x cm are cut from each of the corners and the sides are folded up to make the box.



Which of the following expresses the volume of the box?

a) $V(x) = x(12 + x)(8 + x)$

b) $V(x) = x(12 - x)(8 - x)$

c) $V(x) = x(12 + 2x)(8 + 2x)$

d) $V(x) = x(12 - 2x)(8 - 2x)$

Question 29 — 1 mark

Given that the graph of $f(x)$ contains the point $(-3, 5)$, what point must be on the graph of $f(-x)$?

a) $(-3, -5)$

b) $(3, 5)$

c) $(3, -5)$

d) $(5, -3)$

Question 30 1 mark 122

Determine one positive and one negative coterminal angle with the angle $\frac{5\pi}{6}$.

Question 31 2 marks 123

Evaluate:

$$\left(\sin\frac{11\pi}{3}\right)\left(\sec\frac{11\pi}{6}\right)$$

Question 32 1 mark 124

Given the equation $2\sin^2\theta - 3\sin\theta + 1 = 0$, verify that $\theta = \frac{\pi}{2}$ is a solution.

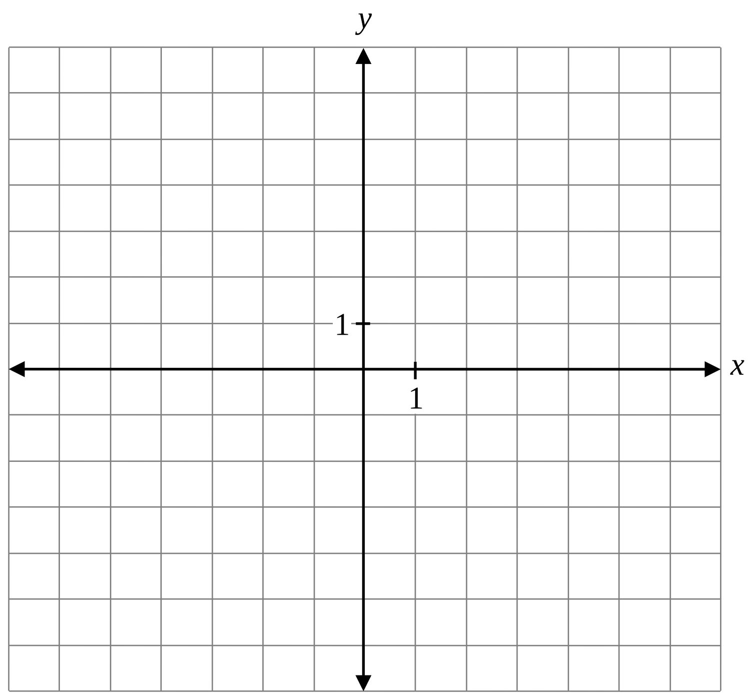
Question 33 2 marks 125

Using the laws of logarithms, expand:

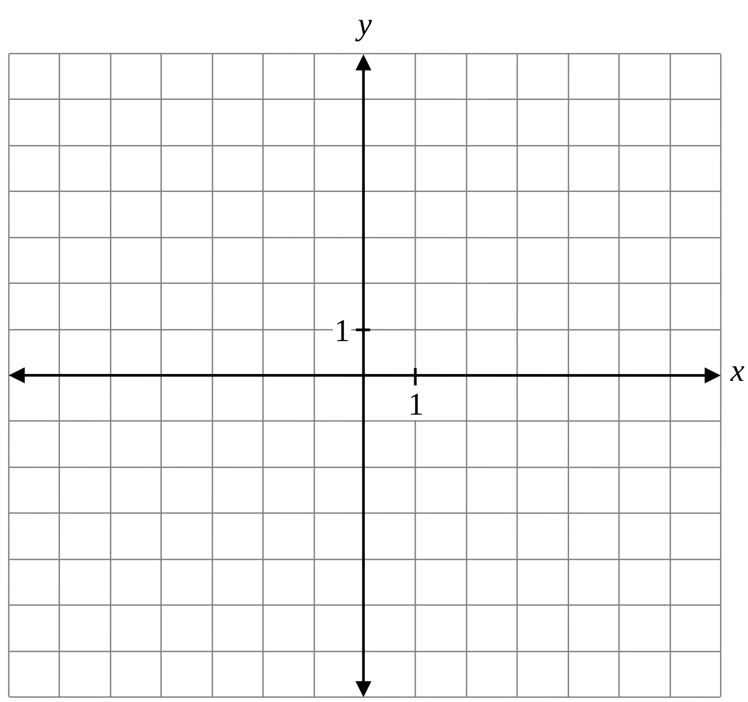
$$\log_a\left(\frac{x \cdot y}{z}\right)$$

Question 34 a) 2 marks b) 1 mark 126 127

a) Sketch the graph of $f(x) = 3^x + 1$.



b) Sketch the graph of $f^{-1}(x)$.



Question 35 2 marks 128

Determine the x-intercept and y-intercept of $y = \log_2(x+4) - 1$.

Question 36 1 mark 129

Explain the error that was made when solving the following equation:

$\sin 2\theta = \cos\theta$, where $\theta \in$ ¡

$$\sin 2\theta = \cos\theta$$

$$2\sin\theta\cos\theta = \cos\theta$$

$$\frac{2\sin\theta\cos\theta}{\cos\theta} = \frac{\cos\theta}{\cos\theta}$$

$$2\sin\theta = 1$$

$$\sin\theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6} + 2k\pi,\ \frac{5\pi}{6} + 2k\pi,\ k \in I$$

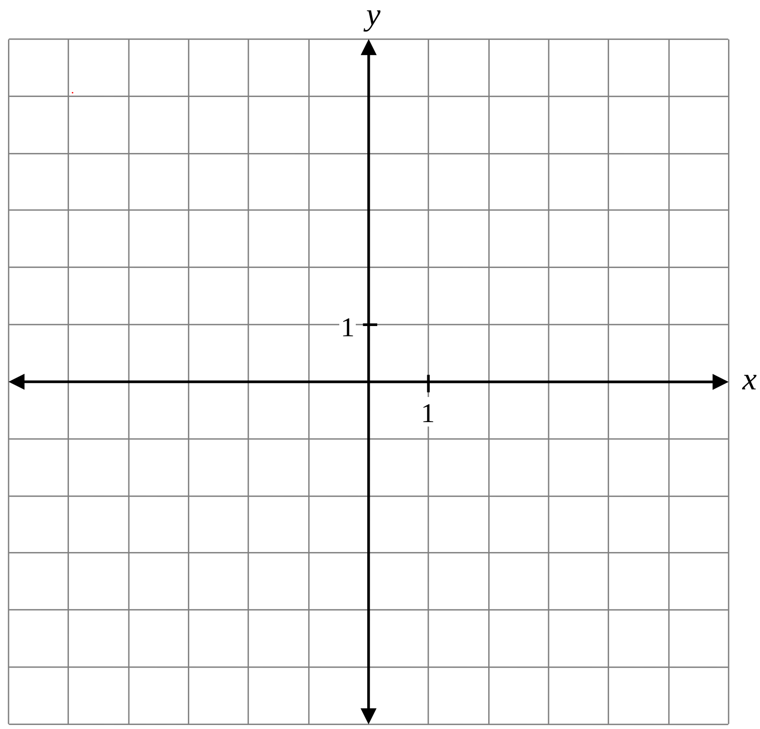
Question 37 a) 1 mark b) 1 mark

130
131

Given $f(x) = x^2 - 2x - 3$ and $g(x) = x + 1$:

a) Write the equation of $y = f(g(x))$.

b) Sketch the graph of $y = f(g(x))$.



Question 38 1 mark 132

Is the point $\left(\frac{3}{4}, -\frac{\sqrt{3}}{4}\right)$ on the unit circle?

Justify your answer.

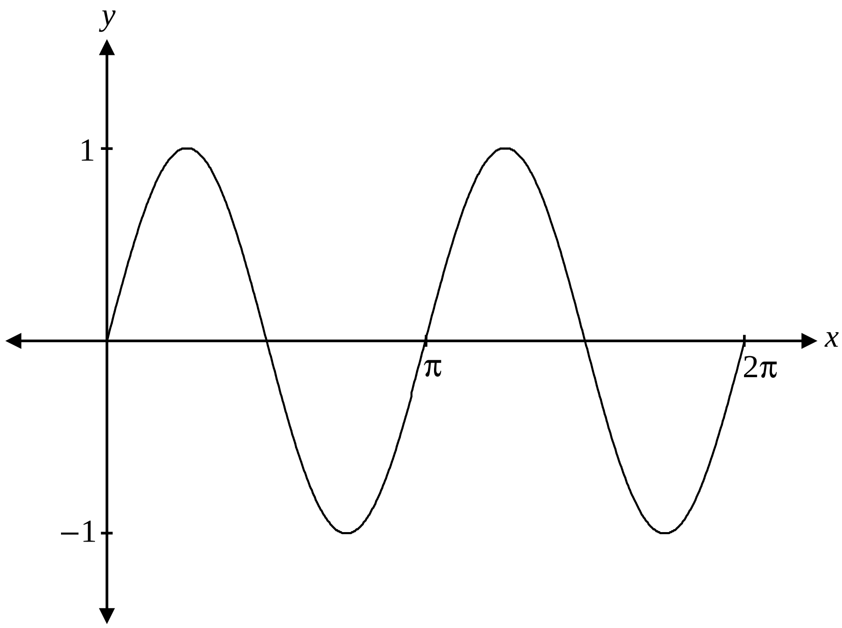
Question 39 1 mark 133

Explain why the equation $\sec\theta = \frac{1}{4}$ has no solution.

Question 40 1 mark 134

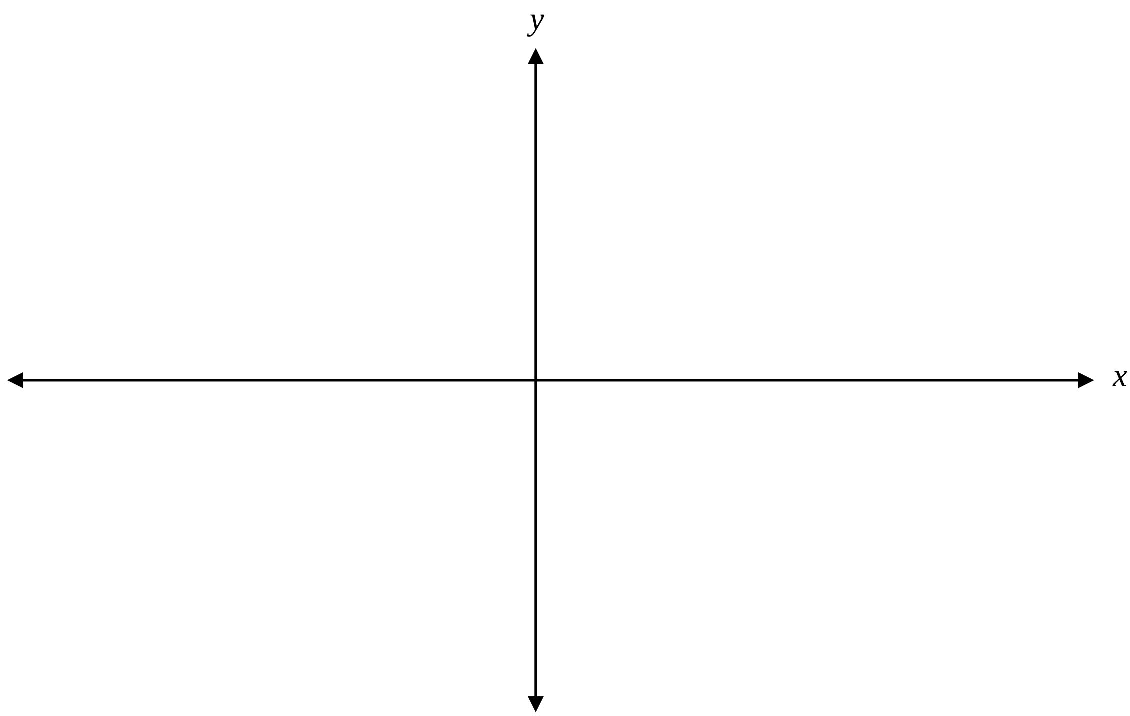
The graph of $y = \sin 2x$ is sketched below.

Explain how to use this graph to solve the equation $\sin 2x = \frac{1}{2}$ over the interval $[0, 2\pi]$.



Question 41 3 marks 135

Sketch the graph of $y = -4\cos(2x)$ over the interval $[-\pi, \pi]$.



Question 42 2 marks 136

Write the equation for $f(x)$ that satisfies all of the following conditions:

- $f(x)$ is a polynomial function of degree 4
- $f(x)$ has a zero at 2 with a multiplicity of 3
- $f(x)$ has a zero at -5
- $f(x)$ has a y-intercept of 80

Question 43 3 marks

Find the exact value of $\sin\left(\frac{19\pi}{12}\right)$.

Question 44 — 4 marks

138

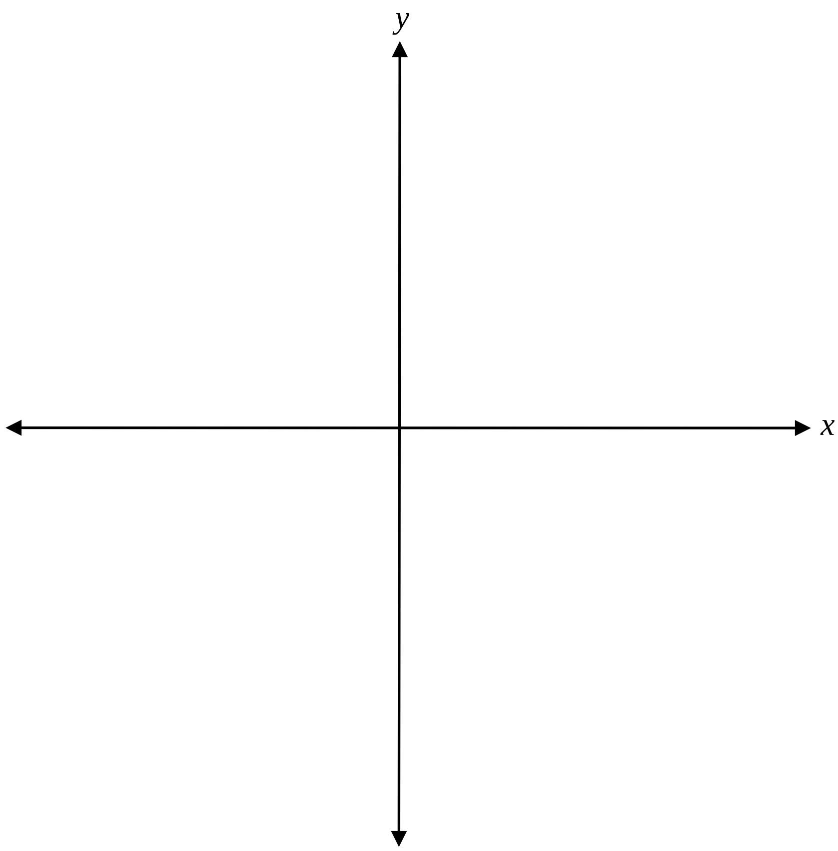
Solve the following equation:

$$2\log_2(x-1) - \log_2(x-5) = \log_2(x+1)$$

Question 45 3 marks 139

Sketch the graph of $f(x) = (x-1)^2(x+2)^3$.

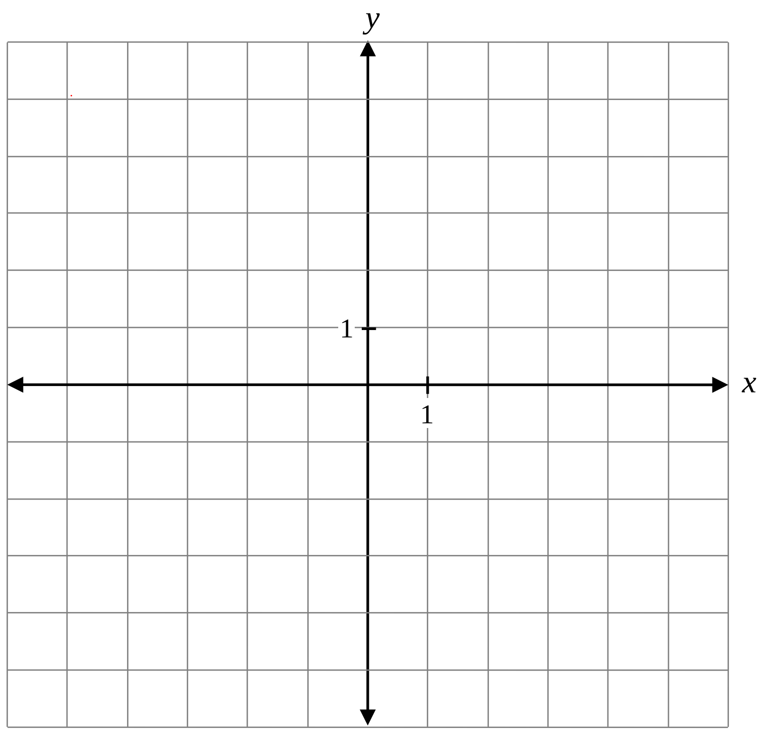
Label the *x*-intercepts and the *y*-intercept.



Question 46 — 4 marks

140

Sketch the graph of $y = -\sqrt{3(x+1)}$.



Question 47 — 3 marks

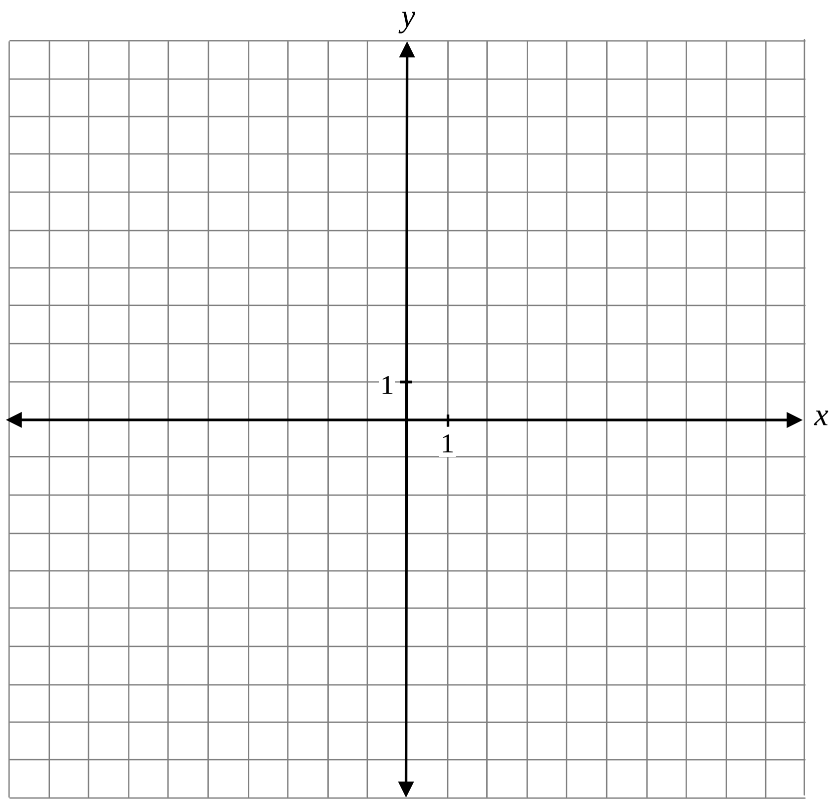
141

Solve:

$${}_{n-1}P_2 = 42$$

Question 48 3 marks

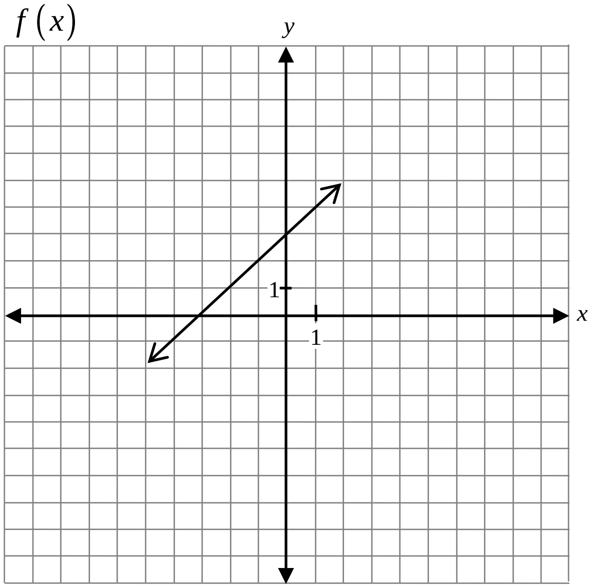
Sketch the graph of $y = \dfrac{2x}{x+2}$.



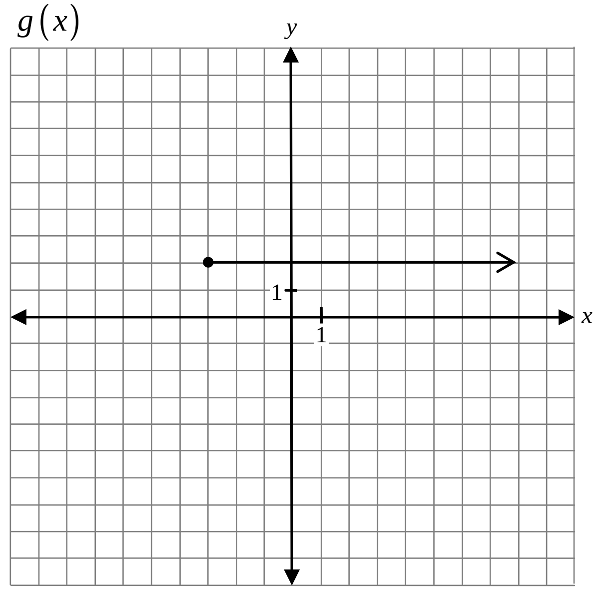
Question 49 2 marks 143

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $(f \cdot g)(x)$.

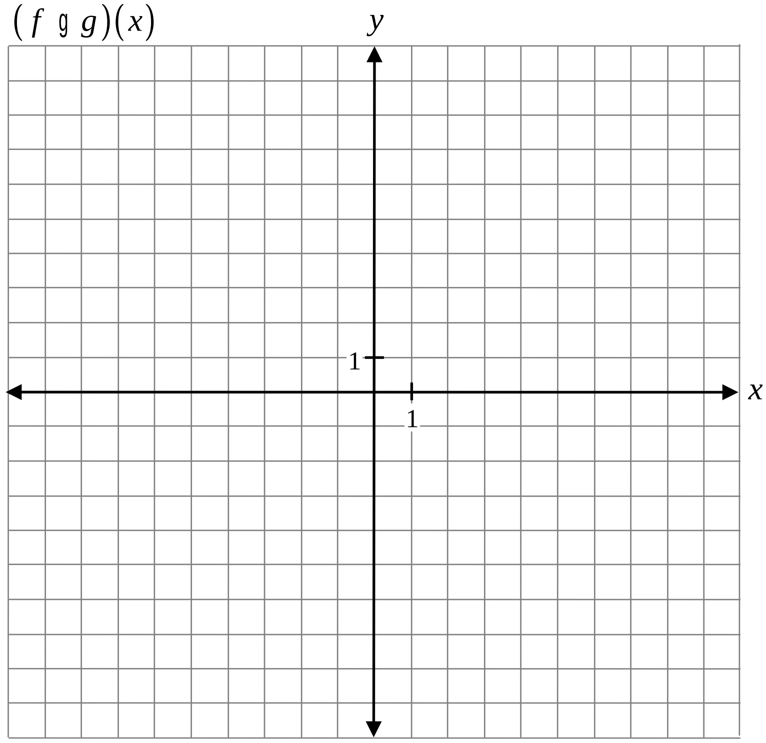
$f(x)$



$g(x)$



$(f \cdot g)(x)$



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